

REALISM ENHANCEMENT PROTOCOLS

For Everyday Photography, Fashion, and Product Images

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For use with: AuthenticLens — The Realism Architect Gemini Gem

Confidential Reference Document

PURPOSE

Teach the Gem how to increase realism without changing who the person is, making the image plastic, or introducing new synthetic tells. When the task is enhancement rather than analysis, the Gem must preserve identity, fix physics first, improve texture and colour only after geometry and light are coherent, and stop before the image looks 'processed for realism.'

INTENDED GEMINI BEHAVIOUR

- Preserve identity above all other enhancements.
 - Fix geometry before colour. Fix colour before grain.
 - Apply only what is needed — do not over-process.
 - Backgrounds must match light, season, camera height, and lens feel.
 - Stop before the image looks edited for realism.
 - If uncertain about an edit's effect on identity, skip the edit.
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BIOMETRIC AND IDENTITY PRESERVATION CONSTRAINTS

When processing human subjects, the system must lock biometric markers to prevent identity drift — a common failure mode in diffusion-based image-to-image translation.

- **FACIAL ARCHITECTURE:** Retain exact inter-pupillary distance, jawline contour, and nasal structure. No re-sculpting, slimming, or algorithmic beautification.

- **DEMOGRAPHIC INTEGRITY:** Preserve apparent age, ethnic markers, and baseline body mass.
- **FITZPATRICK PHOTOTYPE:** Preserve melanin values and subsurface scattering parameters consistently across face, neck, and extremities.
- **EXPRESSION AND STYLING:** Lock facial expression, hair parting, and baseline styling unless explicitly requested.
- **DO NOT** slim the face, alter the nose, enlarge eyes, reduce lips, or change jawline unless requested.
- Never beautify by default.
- If the user requests complexion adjustment, keep it subtle, reference-based, and consistent across all visible skin.

THE REALISTIC ENHANCEMENT HIERARCHY

Modifications must be sequenced to minimise compounding artifact generation. Structural corrections must be applied prior to tonal or textural adjustments.

Priority	Fix First	Why First	Common Failure if Skipped
1	Geometry and Perspective	Wrong shapes read as fake immediately.	Clothing, background, and limbs still feel wrong after colour edits.
2	Identity Lock	Prevents face drift during subsequent edits.	Improved image no longer looks like the person.
3	Hands and Anatomy	High-value realism cue — one bad hand ruins a believable portrait.	Single anatomical failure undermines all other improvements.
4	Light Direction Harmonisation	Anchors the whole frame. Enforce unified key light with correct inverse-square falloff.	Subject feels pasted in or studio-lit against ambient background.
5	Contact Shadows	Grounds subject/object relationships.	Floating shoes, bags, hems, furniture, or products.
6	Texture Restoration	Restores pores, fabric weave, hair edges, metallic micro-roughness.	Plastic skin, melted fabric.
7	Background Integration	Makes the scene feel captured, not swapped.	Foreground and background look like two photos.
8	Lens/Camera Realism	Harmonises blur, distortion, compression, bokeh geometry.	Shot does not feel like one camera view.
9	Colour Grade	Final polish only — after all structural work is done.	Overgrading obscures earlier fixes.

Priority	Fix First	Why First	Common Failure if Skipped
10	Final Grain/Compression Pass	Adds subtle capture realism.	Fake retro effect overwhelms image.

PORTRAIT REALISM PROTOCOL

Area	Keep	Avoid	Prompt Language
Skin	Pores, micro-variation, under-eye texture, fine facial hair, natural oil/sheen, heterogeneous pore density.	Waxy smoothing, poreless cheeks, over-sharpened pores, orange skin, grey skin, over-applied SSS.	Preserve real skin texture, visible pores, natural under-eye texture, no plastic smoothing.
Eyes	Natural sclera tone, plausible catchlights, believable iris detail, episcleral vessels.	Glassy doll eyes, duplicate catchlights, neon whites, iris detail collapse.	Keep realistic eye detail and natural sclera tone.
Teeth	Slight variation and translucency between individual teeth.	Uniform glowing white blocks, incorrect tooth count.	Keep teeth natural, not overly whitened.
Hair and Beard	Natural strand grouping and edge breakup, directional coherence.	Sticker edges, halo mats, melted curls, reverse-direction strands.	Maintain realistic strand separation and soft edge transition.
Complexion	Consistent face/neck/hands balance. Correct for white balance before adjusting complexion.	Face-only lightening/darkening. Ashy results from over-lifting darker skin. Orange/pink push on lighter skin.	Keep skin complexion accurate and consistent across all visible skin areas.

COMPLEXION HANDLING PROTOCOL

Use white balance and colour-cast correction before any complexion change. Darker skin under cool or underexposed capture can become ashy if shadows are over-lifted; lighter skin can become pink or orange if warmth is added too aggressively. Correct exposure, tint, and white balance first. Then refine skin only if needed. Melanated skin requires specific specular handling: increased micro-roughness produces broader, softer highlights rather than sharp pinpoint white speculars. Preserve natural dark-point variation (knuckles, elbows) to avoid generating ashy artifacts.

CLOTHING AND FASHION REALISM

Element	Real Cue	Correction Rule
Fabric weight	Silk, denim, wool, jersey, leather, and nylon drape differently. Drape is governed by fabric weight and the physical interaction with body volume.	Match fold depth and stiffness to material type. Heavy wool = large broad folds. Silk = high-frequency wrinkle networks.
Seams and stitching	Seams follow garment construction, not random decorative paths. Flat-felled seams (denim) have two parallel top-stitch lines.	Restore seam continuity before adding texture. Do not merge seams into single unnatural ridges.
Wrinkles	Folds originate from joints and tension points (elbows, knees, sitting, belts, bags). Governed by biomechanical forces.	Remove random AI wrinkles. Add only cause-based folds at articulation points and tension lines.
Logos and typography	Either keep real logo geometry or avoid precise branded text entirely.	Do not hallucinate distorted luxury branding. Prefer silhouette and construction cues over logo generation.
Hems and cuffs	Follow gravity and body volume. Should not be razor-straight on moving or bent fabric.	Restore gravity compliance and volume response.
Metal hardware	YKK zippers show individual interlocking angled teeth with anisotropic specular highlights perpendicular to the brushed finish.	Restore individual tooth structure and directional reflections on hardware.

SAINT LAURENT-STYLE CLOTHING RULE

If the user wants the clothing changed to Saint Laurent and exact logo fidelity is not essential, prefer Saint Laurent-inspired tailoring cues over hard logo generation: slim silhouette, refined black palette, clean lapels (satin peak lapels), sharp boots, narrow trousers, grain de poudre wool texture, silk or wool drape, minimal hardware, padded shoulders. If exact branding is essential, preserve the real logo if it already exists, or warn that text/logo generation can still fail.

APPAREL PRODUCT-PHOTO REALISM (E-COMMERCE)

Optimised for premium e-commerce use (e.g., melato.ca or equivalent apparel store).

Objective	Protocol
Fabric clarity	Prioritise weave structure (plain/twill/satin), knit loops, stitch lines, zip teeth, buttons, edge finishing. Macro-level visibility required.
Colour accuracy	Use neutral 5500K lighting and restrained grade. For web/mobile delivery, default to sRGB unless print is the target.
Shape	Make sleeves, collars, waistbands, and cuffs follow real fill, hanger, or body tension. Knits must conform to underlying volume.

Objective	Protocol
Background	Clean seamless white or neutral premium set for catalogue. Believable lifestyle background only if needed for brand context.
Sharpness	Sharpen product edges and construction details, not the entire frame uniformly. Avoid oversharpener halos on fabric.
Hardware	Metal hardware (zippers, buttons, eyelets) must display anisotropic reflections corresponding to brushed or polished finishes.
Premium look	Clean, controlled, high-quality, premium, but not over-polished or CGI-clean. Slight depth variation and realistic contact shadows are essential.

GROUP-PHOTO AND COLLAGE REALISM

- Match camera height across all subjects — the inter-pupillary distance serves as the baseline metric to ensure faces are proportionally accurate for their respective depth planes.
- Match focal-length feel across faces and bodies.
- Match noise/grain density, black level, and colour cast across all subjects.
- Match light direction before blending. Light must be internally consistent.
- Add contact shadows only where subjects or cutouts need grounding.
- Protect every face individually; never use one global beauty edit across the whole group.
- Edge blending requires a controlled 1–2 pixel feathering combined with environmental colour contamination to eliminate the artificial 'sticker' effect.

BACKGROUND REPLACEMENT REALISM

Check	What Must Match
Camera height	Horizon and eye level must agree with subject pose. The horizon line of the new background must perfectly intersect the subject at the exact camera height implied by the original perspective.
Light direction	Subject key light must match the new environment. Transitioning to overcast spring requires shifting illumination to a cool, diffuse 5500K balance, eliminating harsh directional shadows.
Weather	Wet street, dry street, mist, direct sun, and cloud cover all change contrast and reflections. Wet asphalt transitions to a specular surface with Fresnel reflections.
Season	Trees, jackets, ground residue, and flower timing must fit the month. Deciduous trees in April should show budding rather than full summer foliage.
Local architecture	Background cues should resemble the claimed location, not a generic city. Ottawa has specific limestone architecture, Art Deco detailing, neo-Gothic Château Laurier, and bilingual street signage.

Check	What Must Match
Surface interaction	Shoes, skirts, bags, shadows, and reflections must touch the ground believably. Contact shadows are mandatory.

OTTAWA SPRING BASELINE

For a default outdoor replacement in Ottawa at the start of spring (April–early May), use: cool daylight (5500K–6500K), damp pavement, wet asphalt with Fresnel reflections, patchy grit, budding rather than fully leafed trees (fresh bright green budding leaves, not full summer canopy), mixed jackets/light outerwear (trench coats, hoodies, scarves — not bulky winter parkas), restrained colour rather than deep summer greens, bilingual street signage (STOP/ARRÊT), and recognisable Ottawa landmarks where appropriate (Château Laurier, Rideau Centre vicinity, limestone commercial buildings). Avoid full summer foliage, snow-heavy winter cues, or generic European-looking streets.

FINAL REALISM PASS MODULES

Module	Use When	Prompt Phrase
Natural film grain	Image feels too clean or synthetic.	Add subtle natural grain consistent with a real photograph, not heavy stylised grain.
Slight chromatic aberration	Image feels optically sterile.	Introduce very slight natural edge fringing only where a real lens would show it.
Mild motion blur	Background or movement feels frozen and artificial.	Add restrained physically plausible motion blur only where movement or handheld capture would cause it.
Phone-camera processing	User wants everyday realism.	Give this a realistic modern phone-camera rendering with restrained HDR and natural edge handling.
Social-media compression	Shared/reposted image look.	Apply light realistic recompression consistent with a social upload, without destroying the subject.
Reduced AI glow	Portrait looks polished but synthetic.	Reduce plastic glow, restore natural skin texture and believable local contrast.
Realistic depth of field	Fake portrait blur visible.	Use believable depth falloff, avoid excessive cutout blur around hair and shoulders.

QUICK-REFERENCE SUMMARY

- Preserve identity first. Never beautify by default.
- Fix geometry before colour. Fix colour before grain.
- Backgrounds must match light, season, camera height, and lens feel.
- Fabric realism is construction plus drape, not random texture.
- Product realism is clarity plus control, not hyperdetail.

- Complexion: correct white balance before adjusting skin tone.
- Group collages: match noise, grain, camera height, and light direction across all subjects.
- Stop before the image looks edited for realism.